Visual Chord Recognition using CNN and Transfer Learning

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Abstract

This paper presents an approach to recognize guitar chords using Convolutional Neural Networks with multiple stages of transfer learning.

1. Introduction

Music recognition has been a long-standing problem. While most solutions focus on audio, visual data can be a valuable resource for describing music, especially in the case of classical instruments like the guitar, piano, or harp.

Our approach centers on utilizing visual data exclusively to address this problem.

1.1. Guitar chords

The task we approached is a classification problem. By a photo of a person holding a guitar chord, the neural network will predict which chord it is. The set of chords is defined as follows: $S = \{$C, D, E, F, A, G, B, Cm, Dm, Em, Fm, Am, Gm, Bm$\}$. At the beginning we will solve the problem only for the subset of S, $M = \{$C, D, E, F, A, G, B$\}, M \subset S$. As difference from major to minor chords is visually very small, the classification task gets much harder when adding minor ones.

1.2. Challenges

There are two main challenges in chord similarity in our classification problem: the hand shape challenge and the hand position challenge.

By holding chords with a hand, different chords can be similar by hand shape, which can be differentiated by only 1 finger, either by its position on the guitar or just by the fact of using it. At the same time, those chords can have the same hand position on the guitar.

Another challenge in the classification is that some chords have identical hand shapes, but they are still different because the hand positions on the guitar are different.

Therefore, we needed to find a solution not just recognizing and classifying hand shapes accurately, but also observing the position of the hand on the guitar. For the 14 main major and minor chords, both types of challenges already exist, especially because the minor counterparts of the same major chords are extremely similar to them.

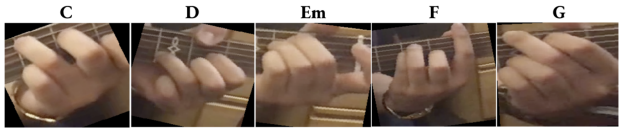

Figure 1. Stanford's team chords post-augmentation. Colors are not normalized here for the clarity of the reader.

2. Related work

A very similar problem was addressed by a Stanford team [4]. Their approach was to firstly extract hand from the image, then classify it with CNN. They classified only 5 different chords and all of them were different in terms of hand shape, see Fig. 1. Because of the chosen chords the position of the hand did not contain any discriminate information, which was not the case in problem chosen by us. Hence, their pipeline did not help us.

3. Method

3.1. Raw data

We collected the majority of the dataset ourselves, capturing images of people holding chords from various distances and orientations. Additionally, open-source images were incorporated to enhance diversity, resulting in a total of 472 images across 14 chords. The dataset was split into training, validation, and test sets (70/15/15 ratio).

3.2. Data augmentation

Firstly, we resized all the images to the size of 400×400. Then we applied different settings of the following augmentations to the images.

Guitarists often hold guitars at various angles, hence, random rotation is the most natural augmentation. We used a range of $\phi, \phi \in [-30°, 30°]$.

Figure 2. Examples of augmented data created by us.

Another augmentation we used was random crop applied with a small setting, resulting in a size of 370×370. This choice was made because when a larger crop is applied to the entire dataset, we encountered the problem of cutting off a hand partially or, in some cases, completely from the images.

The last augmentation we employed was color jitter or grayscale; both of these techniques imitate different colors of guitars and environments, resulting in a more diverse dataset.

We applied three different settings of the discussed augmentations, effectively tripling the resulting volume of the data. Sample augmentations are shown in Figure 2.

We also created a dataset with photos taken of only one person, along with additional augmentation combinations, resulting in around 2000 images in total. This dataset exclusively contained major chords, leading to a total of 7 classes. The application of this dataset will be discussed later in Section 5.1.

To optimize efficiency, we saved all the augmented data, as this proved more efficient than recreating it every time before training.

Prior to training, we resized the data to 160×160 and applied normalization using standard ImageNet [3] parameters (mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$). This step ensured that the pretrained models we used had the same mean and standard deviation of data as during the pretraining process.

4. Experimental results

During experimenting with different settings of training, we made two different datasets out of all the data we gathered.

One was made from photos taken of only one person. Hence, this dataset was not generalized well. We applied a lot of augmentation to the images and got a dataset that consisted of approximately 300 images for each class. Training on this dataset was extremely quick and GoogLeNet [1] reached 96.30% accuracy on the validation set. Because of the way our dataset was made we almost forced the network to pick up the right features. Such training obviously is not reasonable for the final model because the generalization is extremely low.

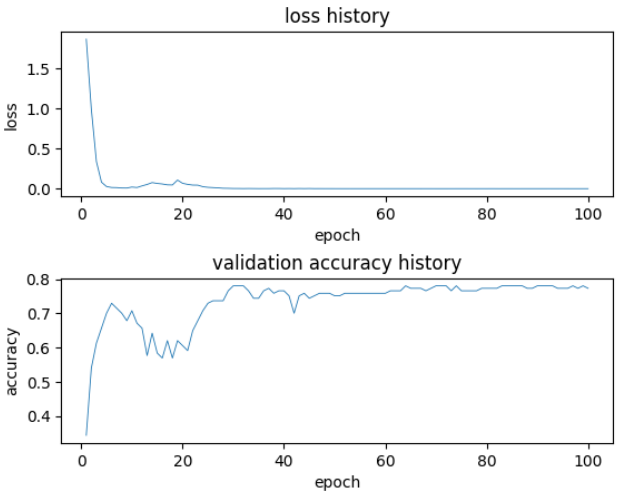

Figure 3. Learning process without transfer learning.

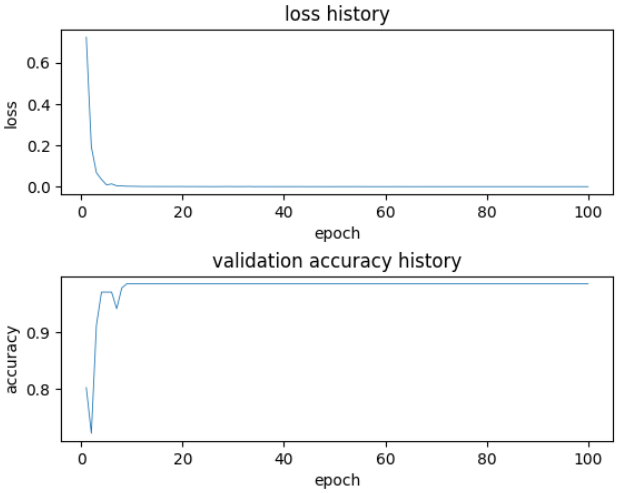

Figure 4. Learning process with transfer learning.

Another setting contained all the data we had and way lower amounts of augmentations. We took photos of 3 different people with different guitars and under different illuminations. The training on this dataset(Fig. 3) was way longer and looking at the validation accuracy history, the network looks lost with such a high variance in the data. The pick accuracy was around 80% which is way worse, comparing to our training on the easier dataset.

We decided to firstly let the network find the related features for our task by training it for 3 epochs on the easy dataset. Then switch to the whole one. Looking at the Fig. 4, there is a clear improvement towards a transfer learning procedure comparing to Fig. 3 without training on the easier data in the beginning.

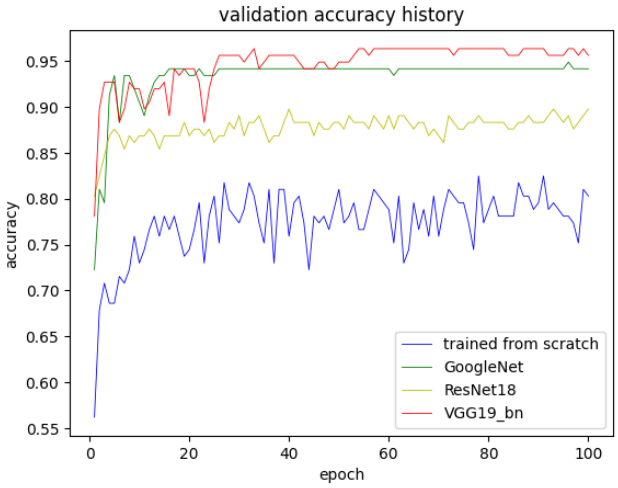

Figure 5. All models were trained on major chords only.

Method	Hard test accuracy
Trained from scratch	22.22%
GoogLeNet	**61.11%**
ResNet18	27.77%
VGG19_bn	44.44%

Table 1. Results

5. Analysis

5.1. Evaluation

Due to the way we created our dataset, the training, the validation and the test sets were very similar to each other, hence, we did not evaluate the model well. To address this problem, the hard test set was created. This set contained images of previously unseen people, clothes and guitars. Which gives a much better understanding of how the network performs in the wild.

5.2. Performance comparison

Looking at the Fig. 5, two best models are VGG19_bn [2] and GoogLeNet [1], where VGG19_bn slightly outperformed GoogLeNet on the validation data. Although VGG19_bn had better validation accuracy, GoogLeNet outperformed VGG19_bn by 16.67% on the hard test(Sec. 5.1).

5.3. Visualization of the first convolution layer

Visualizing the first convolutional layers revealed distinctions between architectures. On Fig. 7 we can see that GoogLeNet mainly focuses on the hand, while according to Fig. 6, VGG's weights are distributed all over the image. Hence, VGG did not pick up the features related to our task and overfit.

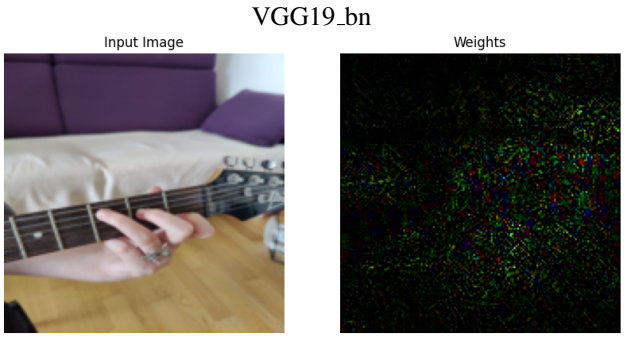

Figure 6. Visualized weights of the first convolution layer of VGG19_bn. Note: weights were scaled by factor of 2 for the visualization.

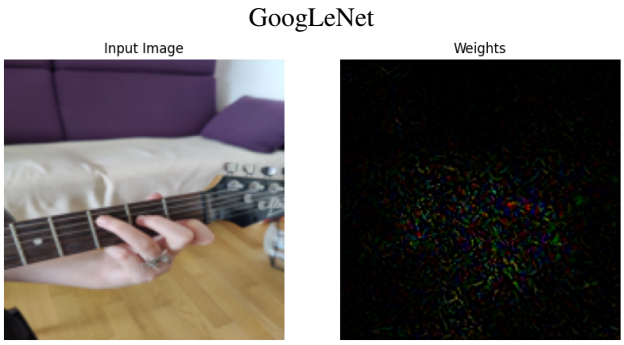

Figure 7. Visualized weights of the first convolution layer of GoogLeNet. Note: weights were scaled by factor of 8 for the visualization.

6. Conclusion

6.1. Generalization

The main problem we encountered during the project was lack of data. Our data did not generalize the network enough as we saw when comparing the accuracy of the default test and the hard test(Sec. 5.1). This problem also came from the similarities between some images as a lot of images had same person, same guitar, same clothes,*etc*.

6.2. Future work

Future work involves enhancing generalization by acquiring more diverse data. Experimentation with alternative architectures like Vision Transformer and exploring strategies for providing complete chord tablature are recommended. The tablature would be a far better outcome, than just chords prediction, because it is a user-friendly notation system, fully describing the way to play the particular composition.

References

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